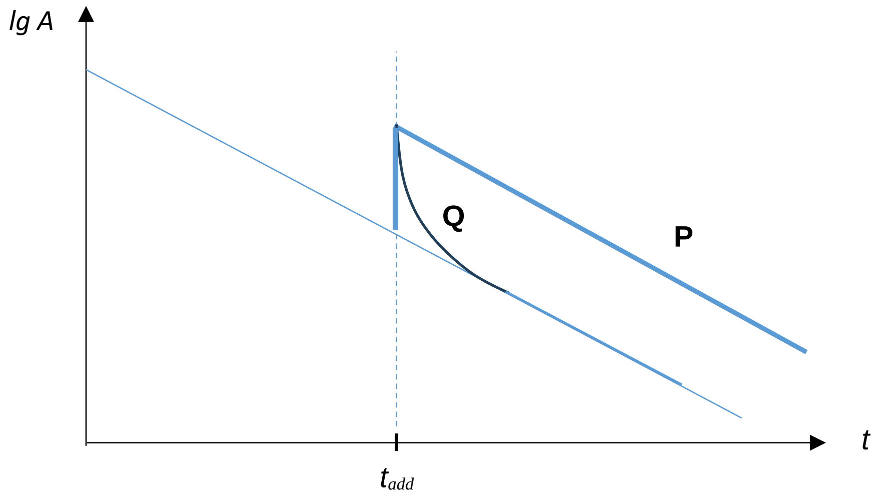
Q7		
(a)	The half-life of a radioactive nuclide is the <u>average time</u> taken for half of the original <u>number</u> of nuclei in a sample of the radioactive nuclide to decay. Or the activity of a sample of the radioactive nuclide to halve.	**B1**
(b)	Activity is the number of disintegrations per unit time.	**B1**
(c)(i)	Half-life = 12.5 h = 12.5 × 60 × 60 = 45000 s Decay constant $= \dfrac{\ln 2}{\text{half-life}} = \dfrac{\ln 2}{45000}$ $= 1.54 \times 10^{-5}\ \text{s}^{-1}$	**B1** **M1** **A1**
(c)(ii)	$A = \lambda N = \dfrac{\ln 2}{T_{1/2}}(0.22N_0) = \dfrac{\ln 2}{(12.5\times 60\times 60)}(0.22N_0)$ $= 3.39\times 10^{-6} N_0$ Bq	**M1** **A1**
(d)(i) and (ii)	 For P, the gradient is the negative of the decay constant. Same nuclide same decay constant and hence same gradient. For Q with a very much shorter half-life compared to K-42, it will approach the original graph quite quickly. ($A = A_{10}e^{-\lambda_1 t} + A_{20}e^{-\lambda_2 t}$. Cannot linearise to give a straight line.)	 **B1** **B1**

Q8		